



Maintenance Schedule

Exhaust gas aftertreatment system, stationary

mtu EGAT - 2x3G-600

mtu EGAT - 2x3G-450

mtu EGAT - 3x3G-600

mtu EGAT - 3x3G-450

mtu EGAT - 3x4G-600

mtu EGAT - 4x4G-600

MS50305/00E



A Rolls-Royce
solution

Product name	Model number	Engine model
mtu EGAT	2x3G-600	8V4000L64FNER, 8V4000L64, 8V4000L32, 8V4000L33, 8V4000L62, 8V4000L63, 12V4000L32F, 12V4000L33, 8V4000L32FB, 8V4000L33FB, 12V4000L32FB
mtu EGAT	2x3G-450	8V4000L64FNER, 8V4000L64, 8V4000L32, 8V4000L33, 8V4000L62, 8V4000L63, 12V4000L32F, 12V4000L33, 8V4000L32FB, 8V4000L33FB, 12V4000L32FB
mtu EGAT	3x3G-600	12V4000L64FNER, 12V4000L64, 12V4000L32F, 12V4000L33, 12V4000L62, 12V4000L63, 16V4000L32F, 16V4000L33, 12V4000L32FB
mtu EGAT	3x3G-450	12V4000L64FNER, 12V4000L64, 12V4000L32F, 12V4000L33, 12V4000L62, 12V4000L63, 16V4000L32F, 16V4000L33, 12V4000L32FB
mtu EGAT	3x4G-600	16V4000L64FNER, 16V4000L64, 16V4000L62, 16V4000L63, 20V4000L32F, 20V4000L33F, 20V4000L32FB
mtu EGAT	4x4G-600	20V4000L64FNER, 20V4000L64, 20V4000L62, 20V4000L63

Table 1: Applicability

Rolls-Royce Solutions refers to Rolls-Royce Solutions GmbH or an affiliated company pursuant to §15 AktG (German Stock Corporation Act) or a controlled company (joint venture), and Rolls-Royce Solutions Ruhstorf GmbH.

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1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product from the manufacturer Rolls-Royce Solutions.

Maintenance concept

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience and therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

Adherence to the specified maintenance intervals is essential to maintain product safety.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.

QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).

QL3: Maintenance work which requires partial disassembly of the product.

QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

Maintenance activities stipulated in any documentation provided by third-parties (suppliers and component manufacturers), particularly those relating to individual components and subsystems, may differ from the information specified in this Maintenance Schedule. The Maintenance Schedule provided by Rolls-Royce Solutions shall take precedence in such case.

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

For unopened RA systems that have been out of service for more than 4 weeks, Rolls-Royce Solutions advises to empty and, if necessary, flush the RA components of the exhaust gas aftertreatment system. Should RA components have been replaced (i.e. the system was opened), Rolls-Royce Solutions recommends immediate draining after extended out-of-service periods.

On-site installations may be subject to other boundary conditions.

The latest version of the specifications is available at <http://www.mtu-solutions.com>.

1.2 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

1.3 Allocation matrix

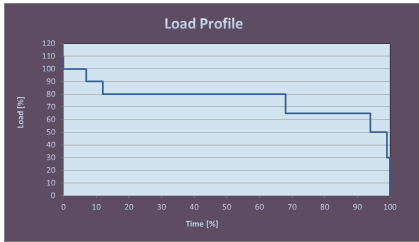
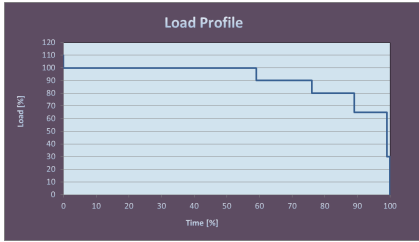
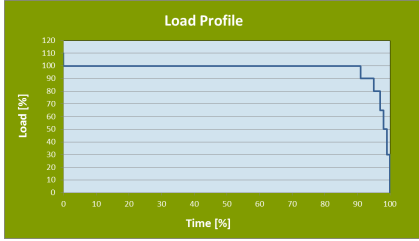
Load profile		Load factor %	Load indicator %	800mg NOx Engine calibration	500mg NOx Engine calibration			
Load %	Time %							
110 ₍₁₀₀₋₁₁₀₎	—	0 to 85	≤ 60	84.000 16.000 (→ 1.7)	84.000 20.000 (→ 1.4)	—	—	
100 ₍₉₀₋₁₀₀₎	7							
90 ₍₈₀₋₉₀₎	5							
80 ₍₆₅₋₈₀₎	56							
65 ₍₅₀₋₆₅₎	26							
50 ₍₃₀₋₅₀₎	5							
30 ₍₅₋₃₀₎	1							
Idle Speed ₍₀₋₅₎	—	86 to 95	≤ 85	84.000 17.500 (→ 1.8)	84.000 21.000 (→ 1.5)	—	—	
110 ₍₁₀₀₋₁₁₀₎	—							
100 ₍₉₀₋₁₀₀₎	59							
90 ₍₈₀₋₉₀₎	17							
80 ₍₆₅₋₈₀₎	13							
65 ₍₅₀₋₆₅₎	10							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	1							
Idle Speed ₍₀₋₅₎	—	96 to 100	≤ 100	84.000 21.000 (→ 1.9)	84.000 24.000 (→ 1.6)	—	—	
110 ₍₁₀₀₋₁₁₀₎	—							
100 ₍₉₀₋₁₀₀₎	91							
90 ₍₈₀₋₉₀₎	4							
80 ₍₆₅₋₈₀₎	2							
65 ₍₅₀₋₆₅₎	1							
50 ₍₃₀₋₅₀₎	1							
30 ₍₅₋₃₀₎	1							
Idle Speed ₍₀₋₅₎	—							

Table 2: Allocation matrix

1.4 Cyclic maintenance tasks with major overhaul at 84000 HRS and intermediate overhaul at 20000 HRS (engine calibration: 500mg NOx)

Qualification	Interval [HRS]	Limit	Item	Maintenance tasks	Option	Task
Exhaust gas aftertreatment						
QL1	-	1 DAY	EXHAUST GAS AFTERTREATMENT SCR mtu	Visually check tanks, lines and dosing system and check for noises	X	WM02039
QL1	-	1 DAY	EXHAUST GAS AFTERTREATMENT SCR mtu	Check plausibility of control variables, measured values, dosing volume and conversion rate in the control system or visualization	X	WM02040
QL1	-	1 WK	EXHAUST GAS AFTERTREATMENT SCR mtu	Visually check reducing agent mixer, catalytic converter housing and tanks	X	WM01655
QL1	-	1 WK	EXHAUST GAS AFTERTREATMENT SCR mtu	Carry out visual inspection of heat insulation	X	WM01657
QL1	-	1 YR	EXHAUST GAS AFTERTREATMENT SCR mtu	Check reducing agent pump	X	WM01659
QL1	4000	5 YR	EXHAUST GAS AFTERTREATMENT SCR mtu	Clean filter in reducing agent tank and before the supply unit, replace if necessary	X	WM01660
QL1	4000	5 YR	EXHAUST GAS AFTERTREATMENT SCR mtu	Replace dosing system filter element	X	WM01661
QL1	6000	6 YR	EXHAUST GAS AFTERTREATMENT SCR mtu	Replace NOx sensor	X	WM01663
QL1	8000	1 YR	EXHAUST GAS AFTERTREATMENT SCR mtu	Check reducing agent day tank, clean if necessary	X	WM01664
QL1	28000	18 YR	EXHAUST GAS AFTERTREATMENT SCR mtu	Check reducing agent pump, replace if necessary	X	WM01666
QL3	20000	18 YR	EXHAUST GAS AFTERTREATMENT SCR mtu	Check emissions, conversion rates, and dosing quantity, replace SCR catalyst elements if necessary	X	WM01668
QL3	20000	18 YR	EXHAUST GAS AFTERTREATMENT SCR mtu	Check emissions, conversion rates, and dosing quantity, replace oxidation catalyst elements if necessary	X	WM01669
QL3	21000	18 YR	EXHAUST GAS AFTERTREATMENT SCR mtu	Replace dosing unit	X	WM01671
QL3	21000	18 YR	EXHAUST GAS AFTERTREATMENT SCR mtu	Replace supply unit	X	WM01672
QL4	84000	18 YR	EXHAUST GAS AFTERTREATMENT SCR mtu	Check exhaust gas aftertreatment system, replace if necessary	X	WM01878
Exhaust gas aftertreatment, third-party supplier						
QL1	-	1 DAY	EXHAUST GAS AFTERTREATMENT SCR	Visually check tanks, lines and dosing system and check for noises	X	WM01673
QL1	-	1 DAY	EXHAUST GAS AFTERTREATMENT SCR	Check filling level of reducing agent tanks	X	WM01674
QL1	-	1 DAY	EXHAUST GAS AFTERTREATMENT SCR	Check plausibility of control variables, measured values, dosing volume and conversion rate in the control system or visualization	X	WM01675
QL1	-	1 WK	EXHAUST GAS AFTERTREATMENT SCR	Check heat insulation for damage	X	WM01677
QL1	-	1 YR	EXHAUST SYSTEM SCR	Visually inspect pressure relief valves	X	WM01678
QL1	-	1 YR	EXHAUST GAS AFTERTREATMENT SCR	Visually check and clean fully integrated system, replace wear parts if necessary	X	WM01683
QL1	4000	5 YR	EXHAUST GAS AFTERTREATMENT SCR	Clean reducing agent and air filter, replace if required	X	WM01679
QL1	4000	5 YR	EXHAUST GAS AFTERTREATMENT SCR	Check operation of sensors with measuring lines and actuator	X	WM01680

DAY = Days
WK = Weeks
YR = Years
HRS = Hours

Qualification	Interval [HRS]	Limit	Item	Maintenance tasks	Option	Task
QL1	4000	5 YR	EXHAUST GAS AFTERTREATMENT SCR	Check and clean reducing agent lines and dosing system	X	WM01681
QL1	6000	6 YR	EXHAUST GAS AFTERTREATMENT SCR	Replace NOx sensor on exhaust gas aftertreatment	X	WM01682
QL3	16000	-	EXHAUST GAS AFTERTREATMENT SCR	Check catalytic converters and conversion rates, replace if necessary	X	WM01685
QL4	84000	18 YR	EXHAUST GAS AFTERTREATMENT SCR	Check exhaust gas aftertreatment system, replace if necessary	X	WM01686

DAY = Days
WK = Weeks
YR = Years
HRS = Hours

1.5 Cyclic maintenance tasks with major overhaul at 84000 HRS and intermediate overhaul at 21000 HRS (engine calibration: 500mg NOx)

Qualification	Interval [HRS]	Limit	Item	Maintenance tasks	Option	Task
Exhaust gas aftertreatment						
QL1	-	1 DAY	EXHAUST GAS AFTERTREATMENT SCR mtu	Visually check tanks, lines and dosing system and check for noises	X	WM02039
QL1	-	1 DAY	EXHAUST GAS AFTERTREATMENT SCR mtu	Check plausibility of control variables, measured values, dosing volume and conversion rate in the control system or visualization	X	WM02040
QL1	-	1 WK	EXHAUST GAS AFTERTREATMENT SCR mtu	Visually check reducing agent mixer, catalytic converter housing and tanks	X	WM01655
QL1	-	1 WK	EXHAUST GAS AFTERTREATMENT SCR mtu	Carry out visual inspection of heat insulation	X	WM01657
QL1	-	1 YR	EXHAUST GAS AFTERTREATMENT SCR mtu	Check reducing agent pump	X	WM01659
QL1	3500	5 YR	EXHAUST GAS AFTERTREATMENT SCR mtu	Clean filter in reducing agent tank and before the supply unit, replace if necessary	X	WM01660
QL1	3500	5 YR	EXHAUST GAS AFTERTREATMENT SCR mtu	Replace dosing system filter element	X	WM01661
QL1	6000	6 YR	EXHAUST GAS AFTERTREATMENT SCR mtu	Replace NOx sensor	X	WM01663
QL1	7000	1 YR	EXHAUST GAS AFTERTREATMENT SCR mtu	Check reducing agent day tank, clean if necessary	X	WM01664
QL1	24500	18 YR	EXHAUST GAS AFTERTREATMENT SCR mtu	Check reducing agent pump, replace if necessary	X	WM01666
QL3	21000	18 YR	EXHAUST GAS AFTERTREATMENT SCR mtu	Check emissions, conversion rates, and dosing quantity, replace SCR catalyst elements if necessary	X	WM01668
QL3	21000	18 YR	EXHAUST GAS AFTERTREATMENT SCR mtu	Check emissions, conversion rates, and dosing quantity, replace oxidation catalyst elements if necessary	X	WM01669
QL3	21000	18 YR	EXHAUST GAS AFTERTREATMENT SCR mtu	Replace dosing unit	X	WM01671
QL3	21000	18 YR	EXHAUST GAS AFTERTREATMENT SCR mtu	Replace supply unit	X	WM01672
QL4	84000	18 YR	EXHAUST GAS AFTERTREATMENT SCR mtu	Check exhaust gas aftertreatment system, replace if necessary	X	WM01878
Exhaust gas aftertreatment, third-party supplier						
QL1	-	1 DAY	EXHAUST GAS AFTERTREATMENT SCR	Visually check tanks, lines and dosing system and check for noises	X	WM01673
QL1	-	1 DAY	EXHAUST GAS AFTERTREATMENT SCR	Check filling level of reducing agent tanks	X	WM01674
QL1	-	1 DAY	EXHAUST GAS AFTERTREATMENT SCR	Check plausibility of control variables, measured values, dosing volume and conversion rate in the control system or visualization	X	WM01675
QL1	-	1 WK	EXHAUST GAS AFTERTREATMENT SCR	Check heat insulation for damage	X	WM01677
QL1	-	1 YR	EXHAUST SYSTEM SCR	Visually inspect pressure relief valves	X	WM01678
QL1	-	1 YR	EXHAUST GAS AFTERTREATMENT SCR	Visually check and clean fully integrated system, replace wear parts if necessary	X	WM01683
QL1	3500	5 YR	EXHAUST GAS AFTERTREATMENT SCR	Clean reducing agent and air filter, replace if required	X	WM01679
QL1	3500	5 YR	EXHAUST GAS AFTERTREATMENT SCR	Check operation of sensors with measuring lines and actuator	X	WM01680

DAY = Days
WK = Weeks
YR = Years
HRS = Hours

Qualification	Interval [HRS]	Limit	Item	Maintenance tasks	Option	Task
QL1	3500	5 YR	EXHAUST GAS AFTERTREATMENT SCR	Check and clean reducing agent lines and dosing system	X	WM01681
QL1	6000	6 YR	EXHAUST GAS AFTERTREATMENT SCR	Replace NOx sensor on exhaust gas aftertreatment	X	WM01682
QL3	16000	-	EXHAUST GAS AFTERTREATMENT SCR	Check catalytic converters and conversion rates, replace if necessary	X	WM01685
QL4	84000	18 YR	EXHAUST GAS AFTERTREATMENT SCR	Check exhaust gas aftertreatment system, replace if necessary	X	WM01686

DAY = Days
WK = Weeks
YR = Years
HRS = Hours